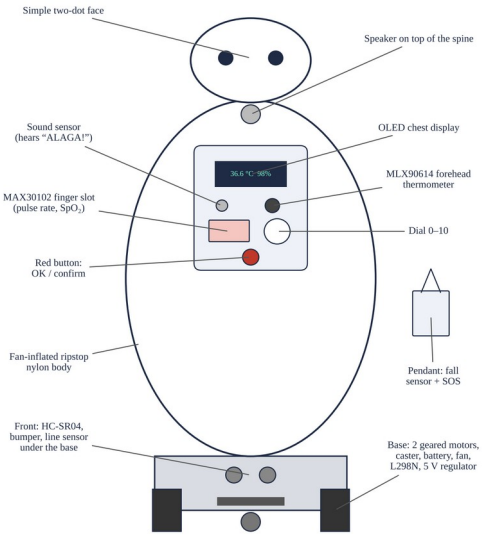
ALAGA

A Low-Cost Wheeled Soft Inflatable Companion Robot for Older Adults Home Alone, with Medication Reminders, Vital-Sign Checks, and Fall Alerts to the Family



Front view, not to scale. About 1.35 m tall when inflated.

ROBOTICS AND INTELLIGENT MACHINES – TEAM (RIM-T)

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Abstract

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In many Filipino homes, older adults stay alone while their children work, and a missed medicine, a fever, or a fall can go unnoticed for hours. Paid caregivers and commercial companion robots cost more than most families can pay. Inspired by the soft healthcare companion of the animated film Big Hero 6, this study built ALAGA, a low-cost wheeled robot with a fan-inflated ripstop nylon body. An ESP32 drives it along a floor-tape route between home stations, stops it before obstacles, reminds the older adult to take medicine and drink water, checks forehead temperature, pulse rate, SpO_2, and pain, plays music and prayers, and sends reports and alerts to the family's phones through Telegram. A pendant with an accelerometer detects falls and has an SOS button. In ____ trips through a mock home, ALAGA reached the correct station in ____% of trips and stopped ____ cm before obstacles. Its readings had mean biases of ____ °C, ____ bpm, and ____% SpO_2 against reference instruments (SpO_2 Arms = ____%). The pendant detected ____% of simulated falls with ____% specificity for daily activities, and alerts reached the family in ____ s. ____% of reminders were on time. Older adults gave a System Usability Scale score of ____ and rated the soft body ____ approachable than a rigid shell (p = ____). The robot and pendant cost PHP ____. A low-cost soft robot can watch over older adults at home and keep their families informed while they work.

Keywords: companion robot; elderly care; soft robotics; fall detection; medication reminder; telemonitoring

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INTRODUCTION

Background of the Study

Filipino families traditionally care for their older members at home, and the law encourages this (Republic Act No. 9994, 2010). Today, however, many adult children work long hours or abroad, so their parents stay home alone during the day. A missed dose of medicine, a fever, or a fall can go unnoticed until someone comes home in the evening.

Falls are a leading cause of injury and death among adults over 60, and lying on the floor for a long time after a fall makes the harm worse (World Health Organization [WHO], 2021). About half of patients with long-term illnesses do not take their medicines as prescribed (WHO, 2003). Older adults who are alone also miss company; robots that talk and play with them have improved mood and reduced stress (Pu et al., 2019; Wada & Shibata, 2007).

The animated film Big Hero 6 (Hall & Williams, 2014) showed a soft, inflatable healthcare companion that checks a person's health, asks for a pain score from 0 to 10, and cannot hurt anyone because its body is air. Real inflatable robot arms have been built for safe contact with people (Sanan et al., 2011), and soft robots are safer around people than rigid ones (Rus & Tolley, 2015). This study asks whether such a companion, on wheels, can be built from low-cost parts to watch over an older adult at home while the family is at work.

Research Gap and Rationale

Commercial companion and telecare robots and paid caregivers cost more than most families can pay. Student-built devices usually do one job, such as a pill reminder, a fall sensor, or a pulse meter, and their accuracy is rarely checked against reference instruments (Bland & Altman, 1986). No low-cost robot found by the researchers joins a soft body that is safe to bump into, simple navigation in a home, medicine and water reminders, vital-sign checks, fall alerts, and family notification in Filipino. It is also not known whether older adults find a soft robot more approachable than a rigid one.

Objectives of the Study

This study aims to design, build, and evaluate ALAGA, a low-cost wheeled soft inflatable companion robot that watches over an older adult at home while the family is at work. Specifically, it seeks to answer the following questions:

1. How reliably does ALAGA travel to each home station (success rate, travel time, and stopping distance from obstacles)?
2. How closely do ALAGA's temperature, pulse rate, and SpO_2 readings agree with reference instruments (mean bias, 95% limits of agreement, mean absolute error, and Arms)?
3. How accurately does the pendant detect simulated falls and ignore daily activities (sensitivity and specificity), and how fast do alerts reach the family?
4. How punctual are the medicine, water, and check-in reminders, and what share of reports and alerts reach the family?
5. How do older adults and family members rate the usability of ALAGA, and is there a significant difference in perceived approachability between its soft body and a rigid shell?
6. How much do ALAGA and its pendant cost to build and to run for one day?

Hypotheses

The following null hypotheses were tested at the 0.05 level of significance:

- $H_0$1: There is no significant difference between ALAGA's readings and the reference readings for temperature, pulse rate, and SpO_2.
- $H_0$2: The pendant's decision (fall or no fall) is not associated with the true event type.
- $H_0$3: There is no significant difference in perceived approachability ratings between the soft inflatable body and the rigid shell.

Significance of the Study

The results of this study will be useful to the following:

Older adults. They get reminders, health checks, company, and a quick way to call for help while alone at home.

Working family members and OFWs. They receive reports and alerts on their phones and can ask for a check-in any time.

Barangay health workers and the OSCA. A low-cost aid for monitoring older adults who live alone during the day.

Future researchers. A tested, open design and an agreement method for low-cost health sensors and fall detection.

The study supports SDG 3 (good health and well-being), SDG 9 (industry, innovation, and infrastructure), and SDG 10 (reduced inequalities).

Scope and Delimitation

This study covers the following:

- One prototype about 1.35 m tall: a two-wheel-drive base with a caster and a fan-inflated ripstop nylon body.
- Navigation along a floor-tape loop with station markers (dock, chair, bed, kitchen), with stops for obstacles by an ultrasonic sensor and a bumper.
- Scheduled medicine, water, and check-in reminders; checks of temperature, pulse rate, SpO_2, and pain; music and prayers; reports and alerts to the family through Telegram over home Wi-Fi.
- A pendant (ESP32-C3 and MPU6050) for fall detection and SOS.
- Tests in a mock home set up in the school, and usability sessions with older adults aged 60 and above, after IRB approval and informed consent.

The study does not cover the following:

- Arms, lifting, feeding, or giving medicine; the older adult takes the medicine and presses a button.
- Diagnosis or treatment; ALAGA reminds, checks, and alerts. It is not a registered medical device.
- Mapping or free roaming; stairs and door sills higher than 1 cm.
- Falls by older adults; falls are simulated only by the researchers on gym mats.
- Homes without Wi-Fi, and continuous use in real homes for weeks.

Definition of Terms

ALAGA. From the Filipino word for care; the robot built in this study.

Companion robot. A robot that keeps a person company, reminds them of daily tasks, and checks on them.

Station. A place in the home marked by a strip of tape across the route: dock, chair, bed, or kitchen.

Line following. Steering by keeping a floor tape under the middle of a row of infrared sensors.

Pendant. A small device worn on a cord that detects falls and has an SOS button.

Fall detection. Deciding that a fall happened when free fall is followed by an impact and then stillness (Bourke et al., 2007).

ESP-NOW. A direct radio link between ESP32 boards that needs no router.

Telegram bot. An automatic account that sends ALAGA's messages to the family's group chat.

SpO_2. The percentage of hemoglobin carrying oxygen in arterial blood.

Mean bias and limits of agreement. The average difference between ALAGA and the reference, and the range that holds 95% of the differences.

Arms. Root-mean-square error of SpO_2 against the reference; ISO 80601-2-61 accepts 4% or less.

System Usability Scale (SUS). A 10-item questionnaire scored from 0 to 100 (Brooke, 1996).

FRAMEWORK

Review of Related Literature

Soft robots are made of materials that bend and give way, which makes them safer around people than rigid machines (Rus & Tolley, 2015). Inflatable links are light and absorb impact; Sanan et al. (2011) built an inflatable arm so that a robot could touch and help people safely. A body kept full of air by a fan is the simplest form of this idea and costs little to make.

Companion robots help older adults through talk and company (Feil-Seifer & Matarić, 2005). A seal robot living with residents of a care house lowered their stress (Wada & Shibata, 2007), and a meta-analysis found that social robots improved mood and reduced agitation in older adults (Pu et al., 2019). Older adults accept such robots more when they are useful, easy to use, and not frightening (Broadbent et al., 2009); faces that look almost human can feel eerie (Mori et al., 2012), so a simple two-dot face was chosen.

Falls are a leading cause of injury among older adults (WHO, 2021). Bourke et al. (2007) detected falls with a body-worn accelerometer by looking for a drop in acceleration (free fall), then a large peak (impact), then little movement (lying still); simple thresholds gave high sensitivity and specificity. Poor adherence to long-term medicines is common, and reminders are among the simplest ways to improve it (WHO, 2003).

Pulse oximeters shine red and infrared light through a fingertip and compare how much of each is absorbed with each heartbeat (Allen, 2007; Jubran, 2015). The ISO 80601-2-61 standard requires an Arms of 4% or less for SpO_2 from 70% to 100% (International Organization for Standardization [ISO], 2017b). Infrared thermometers estimate temperature from the infrared energy the skin gives off (Fraden, 2016); forehead skin is cooler than the body core, so a calibration offset is needed and clinical thermometers are tested under ISO 80601-2-56 (ISO, 2017a). The 0–10 numeric pain scale is valid and quick for adults (Hawker et al., 2011).

When a new instrument is compared with a reference, correlation is not enough; Bland and Altman (1986) recommend the mean difference and its 95% limits of agreement. Usability can be measured with the 10-item System Usability Scale (Brooke, 1996).

Review of Related Studies

Table 1

Related Studies and Their Relation to the Present Study

Study	What was done	Relation to the present study
Sanan et al. (2011)	Built an inflatable robot arm for safe contact with people.	Supports an air-filled body for ALAGA.
Wada & Shibata (2007)	Placed a seal robot with residents of a care house.	Supports ALAGA's companion role.
Bourke et al. (2007)	Detected falls with accelerometer thresholds.	Basis of the pendant's fall rule.
Bland & Altman (1986)	Proposed limits of agreement for comparing two methods.	Method used to test ALAGA's readings.

Note. Table created by the researchers, 2026.

Synthesis

The literature shows that soft bodies are safer and friendlier, that companion robots help older adults, that falls can be detected with low-cost accelerometers, and that vital-sign sensors must be checked against a reference. No study found by the researchers joins these into one low-cost mobile robot that watches over an older adult at home and informs the family.

Theoretical Framework

The study rests on five ideas, each answering a gap (Table 2): socially assistive robotics and telemonitoring, threshold fall detection, soft robotics with the uncanny valley, sensor physics with agreement analysis, and line following with station markers.

Table 2

Research Gaps and the Theories or Concepts Used to Address Them

Research gap	Description	Theory or concept	How it is used
Alone at home	Family is away at work for most of the day.	Socially assistive robotics; telemonitoring	Reminders, company, reports to the family.
Falls go unnoticed	The older adult may not reach a phone.	Threshold fall detection	Pendant alerts the family in seconds.
Robots can hurt or scare	A rigid moving machine in a small home.	Soft robotics; uncanny valley	Air body, slow speed, plain face.
Cheap sensors may be wrong	Low-cost modules are rarely validated.	PPG, IR thermometry, agreement analysis	Calibrate, then test with Bland–Altman.
Moving in a home cheaply	Mapping needs costly sensors.	Line following with markers	Tape route between stations.

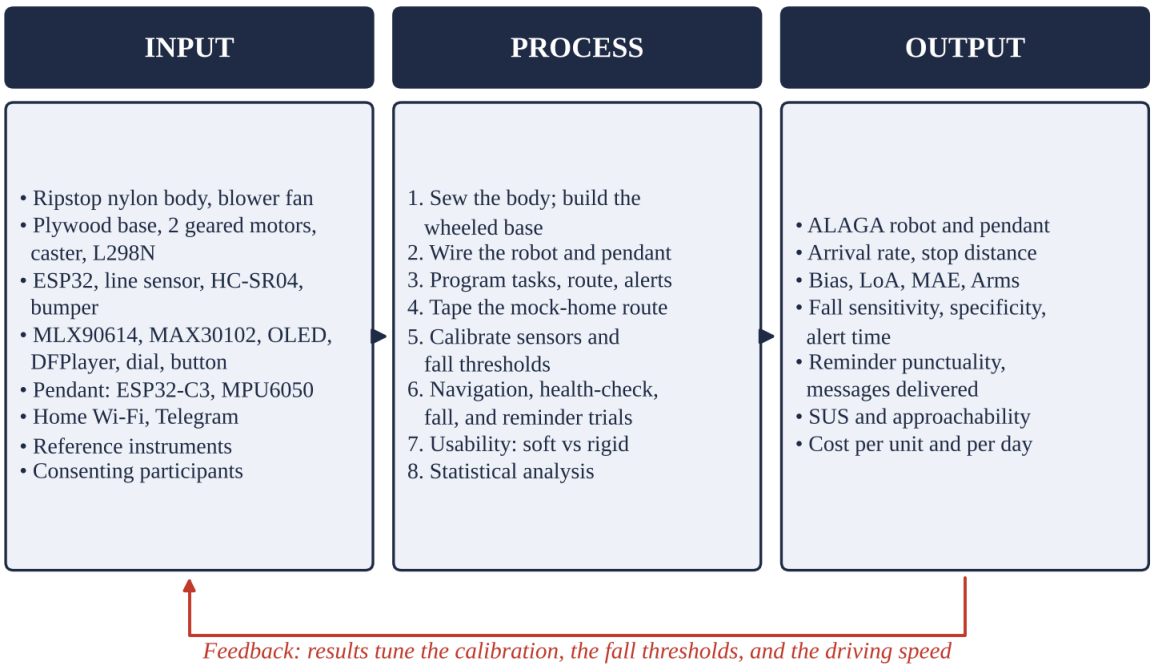
Note. Table created by the researchers, 2026.

Conceptual Framework

The study follows the input–process–output model in Figure 1. Results are fed back to tune the calibration offsets, the fall thresholds, and the driving speed.

Figure 1

Conceptual Framework of the Study in Input–Process–Output Form



Note. Figure created by the researchers, 2026.

METHODOLOGY

Research Design

This is a quantitative developmental study with method-comparison, experimental, and within-subjects usability tests. The robot and pendant are built and calibrated first; then navigation, vital-sign agreement, fall detection, reminders, and alerts are tested in a mock home, and older adults and family members rate the robot with a soft body and with a rigid shell. Every value comes from the researchers' own tests.

Table 3

Variables of the Study

Independent (varied)	Dependent (measured)	Controlled (held fixed)
Target station; obstacle placed	Success rate, travel time, stop distance	Tape route, floor, speed 0.2 m/s
Measurement method (ALAGA; reference)	Bias, LoA, MAE, Arms	Room 26–30 °C; 5-min rest
Event type (fall; daily activity)	Sensitivity, specificity, alert time	Pendant worn at the chest
Body type (soft; rigid)	Punctuality; SUS; approachability (1–5)	Same prompts; order alternated

Note. Table created by the researchers, 2026.

Materials and Equipment

Components were bought from local electronics shops and online stores. The reference forehead thermometer and pulse oximeter were borrowed from the school clinic. Expenses are in Appendix A.

Table 4

Materials and Components of ALAGA

Material / Component	Quantity	Purpose
White 70D ripstop nylon, seam tape, zipper	3 m; 1 set	Inflatable body and head
12 V DC blower fan with finger guard	1	Keeps the body inflated
12 V geared motors with 65-mm wheels; caster	2; 1	Two-wheel-drive base
L298N driver; 5-channel line sensor; bumper	1 each	Driving and route following
ESP32 development board	1	Controller, Wi-Fi, ESP-NOW
MLX90614; MAX30102	1 each	Temperature, pulse rate, SpO_2
HC-SR04 ultrasonic sensor	1	Obstacle detection
OLED; DFPlayer Mini, microSD, 3 W speaker	1 set	Display and voice

KY-037 sound sensor, button, 10 kΩ dial	1 each	Calling, confirming, choosing
12 V 7 Ah battery, LM2596, D4184, fuse	1 set	Power and fan switch
Pendant: ESP32-C3, MPU6050, buzzer, LiPo	1 set	Fall detection and SOS
Reference thermometer and pulse oximeter	1 each	Reference readings (borrowed)

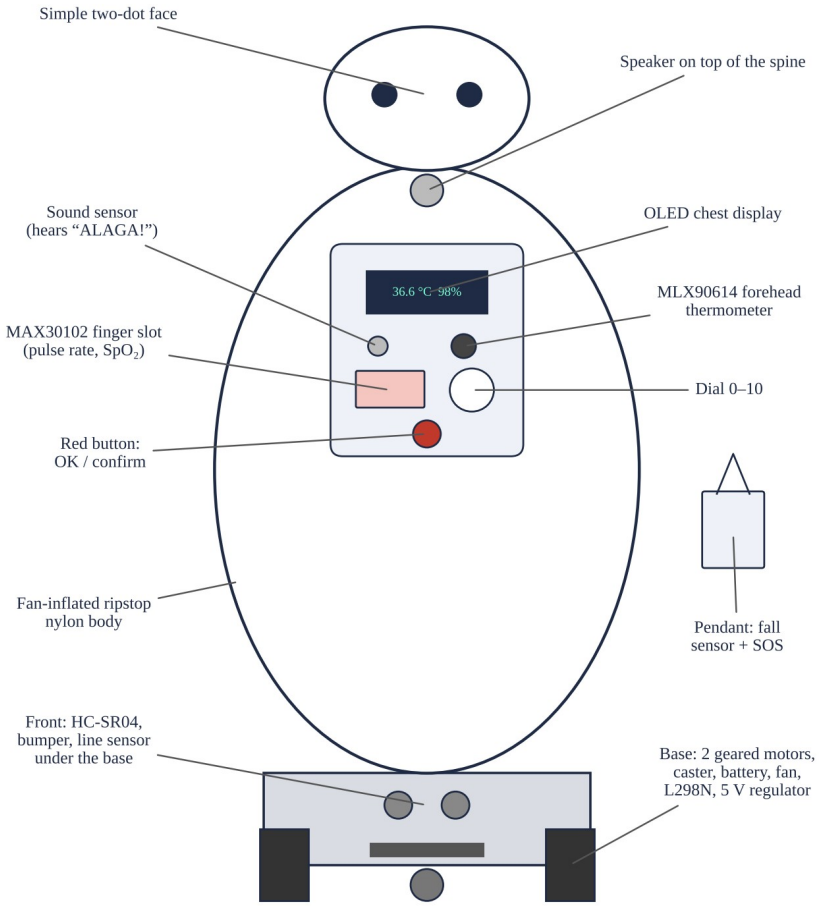
Note. Table created by the researchers, 2026.

Specification of the Device

- Body: white 70D ripstop nylon, about 1.1 m tall on a 0.2-m base (1.35 m in all), kept inflated by a 12 V blower fan (Figure 2).
- Base: 40 × 40 cm plywood box; two 12 V geared motors with 65-mm wheels at the back and a swivel caster at the front; top speed limited to about 0.2 m/s.
- Front sensors: 5-channel infrared line sensor under the base, HC-SR04 ultrasonic sensor, and a bumper with two microswitches.
- Chest box at about 85 cm: MLX90614 thermometer, MAX30102 finger slot, 0–10 dial, red button, OLED display, sound sensor, and speaker.
- Controller: ESP32 with Wi-Fi (Telegram messages) and ESP-NOW (pendant); 12 V 7 Ah sealed lead-acid battery with a 5 V regulator.
- Pendant: ESP32-C3, MPU6050 accelerometer, buzzer, button, and a 400 mAh battery on a neck cord.

Figure 2

Design and Parts of ALAGA



Front view, not to scale. About 1.35 m tall when inflated.

Note. Figure created by the researchers, 2026.

Construction and Preparation

Inflatable Body

The body was cut from ripstop nylon in front, back, and side panels, sewn with flat seams, and joined to a separately sewn head by a neck tube. It is a constant-air inflatable: the fan runs during every task and small leaks at the seams are expected. A PVC pipe spine inside the body holds the chest box at the front.

Mobile Base

The motors were bolted with brackets to the rear corners of the plywood base and the caster at the front center, giving three points of contact. The heavy battery sits low between the

wheels, so the light body cannot tip the robot at low speed. The line sensor is mounted 1 cm above the floor in front of the caster, with the ultrasonic sensor and bumper above it.

Electronics and Pendant

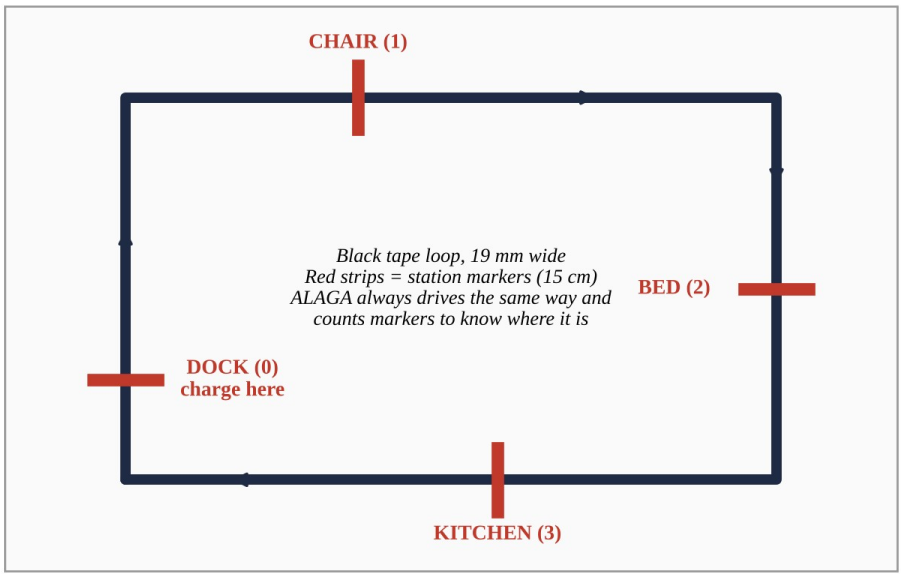
The MLX90614, MAX30102, and OLED share one I²C bus at 100 kHz. An L298N drives the motors, and a D4184 logic-level MOSFET module switches the fan. The pendant reads the MPU6050 at 50 Hz and sends fall, SOS, and once-a-minute status messages to the robot by ESP-NOW. Pin connections are in Appendix G.

Home Route

Black electrical tape (19 mm) was laid on a light floor as one closed loop through the stations (Figure 3). A 15-cm strip across the loop marks each station. The robot counts markers in travel order, so it always knows which station it has reached and returns to the dock by going around the loop.

Figure 3

Example Home Route With Station Markers



Note. Figure created by the researchers, 2026.

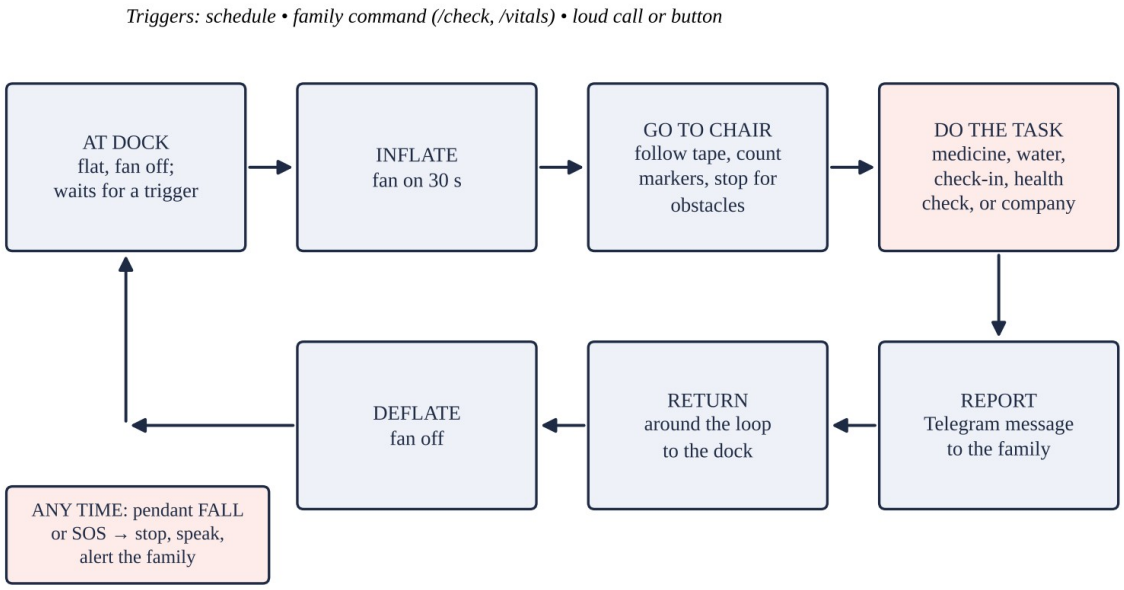
Control Logic

The program follows Figure 4. ALAGA waits flat at the dock. When a scheduled task is due, when the family sends a Telegram command, or when the older adult calls out or presses the chest button, the fan inflates the body and ALAGA drives to the chair, stopping and saying

“Excuse me” if something is in the way. It then gives the medicine, water, or check-in reminder, runs a health check, or keeps company with music and prayers. Each result is sent to the family, and ALAGA returns to the dock and deflates. A fall or SOS message from the pendant interrupts any task and is sent to the family at once.

Figure 4

Control Logic of ALAGA



Note. Figure created by the researchers, 2026.

Table 5

Screening Limits and Triage Levels

Measure	GREEN (normal)	YELLOW (see health worker)	RED (see health worker now)
Temperature (°C)	below 37.5	37.5 to 38.9	39.0 or higher
SpO_2 (%)	95 to 100	92 to 94	below 92
Pulse rate at rest (bpm)	60 to 100	40–59 or 101–130	below 40 or above 130
Pain (0–10 NRS)	0 to 3	4 to 6	7 to 10

Note. Table created by the researchers, 2026.

Table 6

Messages Sent to the Family Through Telegram

Event	Example message
Health check	GREEN: Lola's check at 07:04: 36.6 C, PR 78, SpO2 97%, pain 1.
Medicine	Lola confirmed the 12:30 medicine at 12:33. / “… did not confirm …”

Check-in	Lola answered the 15:00 check-in: OK. / “No answer …”
Fall or SOS	FALL detected on Lola's pendant at 10:42. Please call Lola now.
Family commands	/status, /check, /vitals, /water

Note. Table created by the researchers, 2026.

Calibration

Forehead skin temperature from the MLX90614 was converted to body temperature with a linear equation fitted to readings from the first 10 volunteers (calibration set). Constant offsets for pulse rate and SpO_2 were found from the same set. These volunteers were not used again in the agreement analysis. The fall thresholds were set from the first 10 simulated falls and daily activities and then kept fixed.

Experimental Setup

A mock home of about 4 × 5 m was set up in a classroom with a chair, a bed, a table, and a kitchen corner. Health checks were done at a room temperature of 26 to 30 °C after 5 minutes of seated rest. Simulated falls were done by the researchers onto 20-cm gym mats while wearing the pendant at the chest.

Table 7

Trials Conducted in the Study

Trial set	What is measured	Number	Reference
Navigation	Arrival, travel time, obstacle stops	____ trips; 30 obstacles	Stopwatch, tape
Bench temperature	IR reading of a 35.0–40.0 °C water-bath target	6 × 10	Probe thermometer
Calibration set	Temperature, PR, SpO_2	10 volunteers × 3	Clinic instruments
Validation set	Temperature, PR, SpO_2	____ volunteers × 3	Clinic instruments
Falls and activities	Pendant decision, alert time	____ falls; ____ activities	Video
Reminders, messages	On-time reminders; delivered messages	5 days × 7 tasks	Phone time stamps
Usability	SUS; approachability, soft vs rigid	____ older adults; ____ family	Counterbalanced

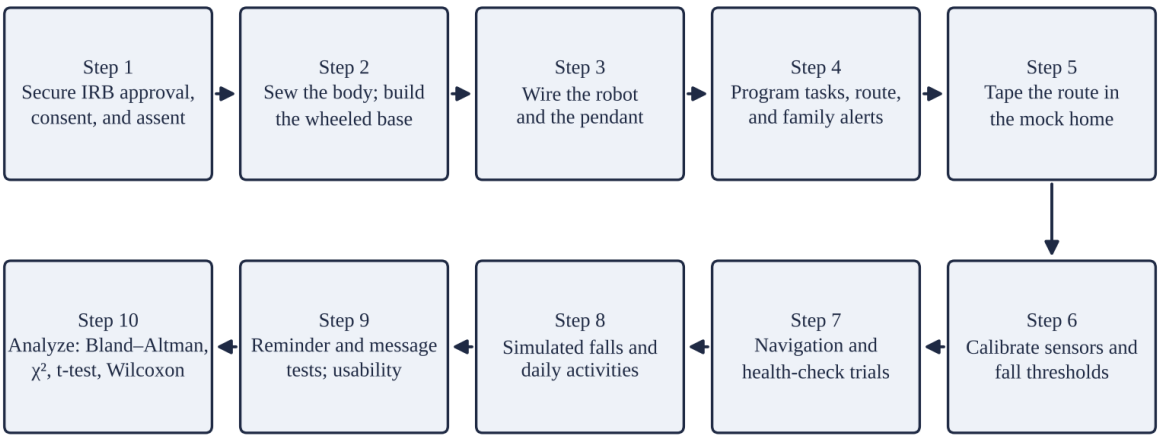
Note. Table created by the researchers, 2026.

Data Gathering Procedure

The ten steps in Figure 5 were followed. Each participant received a code number, and no names were written on the data sheets (Appendix B).

Figure 5

Flowchart of the Research Procedure



Note. Figure created by the researchers, 2026.

Statistical Treatment

The following formulas were used:

Success rate = correct arrivals ÷ trips × 100	where navigation reliability (%)	(1)
d = ALAGA − reference; Bias = Σd ÷ n	where difference and mean bias	(2)
LoA = Bias ± 1.96 × SD(d); Arms = √(Σd² ÷ n)	where 95% limits of agreement; SpO_2 RMS error	(3)
Sensitivity = TP ÷ (TP + FN); Specificity = TN ÷ (TN + FP)	where falls detected; daily activities correctly ignored	(4)
t_alert = t_phone − t_event	where alert time (s)	(5)
Punctuality = on-time reminders ÷ reminders × 100	where reminders within 1 min of schedule (%)	(6)
SUS = 2.5 × [Σ(odd item − 1) + Σ(5 − even item)]	where System Usability Scale score (0–100)	(7)

Agreement was shown with Bland–Altman plots, and $H_0$1 was tested with a paired t-test for each measure. Pendant decisions were summarized in a confusion matrix and $H_0$2 was tested with a chi-square test of independence. Approachability ratings (1–5) for the soft body and the rigid shell were compared with the Wilcoxon signed-rank test ($H_0$3). The level of significance was 0.05, and Microsoft Excel and JASP were used.

Safety and Ethical Considerations

This study involves human participants, including older adults, so it followed the ISEF rules for human participants research used by the NSTF. Before any testing:

- The research plan was reviewed and approved by the school's Institutional Review Board (IRB).
- Each participant signed an informed consent form; minors also gave assent with parental consent (Appendix E).
- Older adults only did seated, non-invasive health checks and usability sessions; they never simulated falls.
- Falls were simulated only by the researchers onto 20-cm gym mats, with knee and elbow pads, under adult supervision.
- Telegram messages went only to a private group of the family and the researchers; data were coded and deleted after the fair.
- All sources are cited in APA 7th edition; no data were fabricated.

Physical risks were controlled as listed in Appendix D: a top speed of 0.2 m/s, a soft body and bumper, a finger guard on the fan, a fuse on the battery, and wiping of the sensors with 70% alcohol between participants.

FINDINGS

Results and Discussion

[NEED: every ____ in this chapter must be replaced with values the researchers measured. No value may be entered that was not measured.]

Results for Objective 1: Navigation

Table 8 shows the navigation trials. ALAGA reached the correct station in ____ of ____ trips (____%), with a mean travel time of ____ s to the chair. It stopped ____ cm (SD ____) before obstacles and touched an obstacle ____ times.

Table 8

Navigation Results in the Mock Home

Target	Trips	Correct arrivals (%)	Travel time, mean ± SD (s)	Stop distance (cm)
Chair	____	____	____ ± ____	____
Bed	____	____	____ ± ____	____
Kitchen	____	____	____ ± ____	____
Dock	____	____	____ ± ____	____

Note. Table created by the researchers, 2026.

Results for Objective 2: Agreement of the Health Check

Table 9 shows the infrared thermometer against the water-bath target, and Table 10 the agreement with the reference instruments. Temperature had a bias of ____ °C (LoA ____ to ____), pulse rate ____ bpm, and SpO_2 ____% with an Arms of ____%, which ____ the 4% limit of ISO 80601-2-61. The paired t-test gave p = ____, ____, and ____, so $H_0$1 is ____.

Table 9

Bench Test of the Infrared Thermometer Against a Water-Bath Target

Target (°C)	Probe reading (°C)	ALAGA mean ± SD (°C)	Error (°C)
35.0	____	____ ± ____	____
36.0	____	____ ± ____	____
37.0	____	____ ± ____	____
37.5	____	____ ± ____	____
38.0	____	____ ± ____	____
40.0	____	____ ± ____	____

Note. Table created by the researchers, 2026.

Table 10

Agreement of ALAGA With the Reference Instruments (Validation Set)

Measure	n	Bias	95% LoA	MAE	Arms	p
Temperature (°C)	____	____	____ to ____	____	____	____

Pulse rate (bpm)	____	____	____ to ____	____	____	____
SpO_2 (%)	____	____	____ to ____	____	____	____

Note. Table created by the researchers, 2026.

Figure 6

Bland–Altman Plots of Temperature, Pulse Rate, and SpO_2

[NEED: Bland–Altman plots from the measured validation data: difference (ALAGA − reference) against the mean of the two, with lines for the bias and the 95% limits of agreement.]

Note. Figure created by the researchers, 2026.

Results for Objective 3: Fall Detection and Alerts

The pendant detected ____ of ____ simulated falls (sensitivity ____%) and correctly ignored ____ of ____ daily activities (specificity ____%); χ^2 = ____, p = ____, so $H_0$2 is ____ (Table 11). Fall and SOS alerts reached the family's phones in a mean of ____ s.

Table 11

Confusion Matrix of Pendant Decisions

	True fall	Daily activity	Total
Pendant: fall	TP = ____	FP = ____	____
Pendant: no fall	FN = ____	TN = ____	____
Total	____	____	____

Note. Table created by the researchers, 2026.

Results for Objective 4: Reminders and Messages

Over ____ days, ____% of reminders started within 1 minute of schedule, and ____% of reports and alerts reached the family (Table 12).

Table 12

Reminders and Messages Over the Test Days

Measure	Count	Result
Reminders on time (within 1 min)	____ / ____	____%
Medicine reminders confirmed	____ / ____	____%
Reports and alerts delivered	____ / ____	____%
Alert time, fall or SOS (s)	____	mean ____

Note. Table created by the researchers, 2026.

Results for Objective 5: Usability and Approachability

Older adults gave a mean SUS score of ____ and family members ____. Approachability was rated ____ for the soft body and ____ for the rigid shell (Wilcoxon z = ____, p = ____), so $H_0$3 is ____ (Table 13).

Table 13

Usability and Approachability Ratings

Measure	Soft body	Rigid shell	Test	p
SUS, older adults (0–100)	____	—	Mean ± SD	—
SUS, family members (0–100)	____	—	Mean ± SD	—
Approachability (1–5), median	____	____	Wilcoxon z = ____	____

Note. Table created by the researchers, 2026.

Results for Objective 6: Cost

The robot and pendant cost PHP ____ to build and PHP ____ per day to run (Table 14).

Table 14

Cost of Building and Running ALAGA

Item	Amount
Robot and pendant parts (Appendix A)	PHP ____
Energy per day (Wh)	____
Electricity per day at PHP ____/kWh	PHP ____
Days per battery charge	____

Note. Table created by the researchers, 2026.

Summary of Results by Objective

Table 15

Summary of Results by Objective

Objective	Result	Decision
1. Navigation	____% correct arrivals; stop at ____ cm	—
2. Agreement	Bias ____ °C, ____ bpm, ____ %; Arms ____%	$H_0$1 ____
3. Falls and alerts	Sens. ____%, spec. ____%; ____ s	$H_0$2 ____
4. Reminders	____% on time; ____% delivered	—
5. Usability	SUS ____; soft vs rigid p = ____	$H_0$3 ____
6. Cost	PHP ____ per unit; PHP ____ per day	—

Note. Table created by the researchers, 2026.

CONCLUSIONS

Assessment of the Findings

Based on the results of the study, the following conclusions were drawn:

1. ALAGA was built as a wheeled, fan-inflated soft robot with reminders, health checks, company, and family messages, together with a fall and SOS pendant.
2. It reached the correct station in ____% of trips and stopped ____ cm before obstacles.
3. Its readings had biases of ____ °C, ____ bpm, and ____% SpO_2, with an SpO_2 Arms of ____%; $H_0$1 is ____.
4. The pendant reached ____% sensitivity and ____% specificity, and alerts arrived in ____ s; $H_0$2 is ____.
5. ____% of reminders were on time, and ____% of messages reached the family.
6. Older adults gave a SUS score of ____ and rated the soft body ____ approachable (p = ____); $H_0$3 is ____.
7. The robot and pendant cost PHP ____ to build and PHP ____ per day to run.

Limitations

ALAGA was tested in a mock home, not in real homes over weeks. Falls were simulated by young researchers, whose falls may differ from those of older adults. A button press confirms that a reminder was heard, not that the medicine was taken. Low-cost pulse oximeter sensors can read higher in people with darker skin and cold fingers. Messages need home Wi-Fi, and the robot cannot find its way off the tape route.

Applications

Families whose older members stay home alone during working hours can use ALAGA for reminders, company, and alerts. Barangay health workers and the Office for Senior Citizens Affairs can lend units to older adults who live alone.

RECOMMENDATIONS

Based on the findings and limitations of the study, the researchers recommend the following:

1. Run a two-week pilot in real homes with older adults who stay alone, with family and IRB consent.
2. Add an automatic charging dock and battery-level reports.
3. Add a 4G modem for homes without Wi-Fi, and a pill box that opens only at the scheduled time.
4. Test the fall pendant with data from real falls of older adults in partnership with a hospital.
5. Add prompts in Bikol and simple voice commands.

REFERENCES

Allen, J. (2007). Photoplethysmography and its application in clinical physiological measurement. Physiological Measurement, 28(3), R1–R39. https://doi.org/10.1088/0967-3334/28/3/R01

Bland, J. M., & Altman, D. G. (1986). Statistical methods for assessing agreement between two methods of clinical measurement. The Lancet, 327(8476), 307–310. https://doi.org/10.1016/S0140-6736(86)90837-8

Bourke, A. K., O'Brien, J. V., & Lyons, G. M. (2007). Evaluation of a threshold-based tri-axial accelerometer fall detection algorithm. Gait & Posture, 26(2), 194–199. https://doi.org/10.1016/j.gaitpost.2006.09.012

Broadbent, E., Stafford, R., & MacDonald, B. (2009). Acceptance of healthcare robots for the older population: Review and future directions. International Journal of Social Robotics, 1(4), 319–330. https://doi.org/10.1007/s12369-009-0030-6

Brooke, J. (1996). SUS: A “quick and dirty” usability scale. In P. W. Jordan, B. Thomas, B. A. Weerdmeester, & I. L. McClelland (Eds.), Usability evaluation in industry (pp. 189–194). Taylor & Francis.

Espressif Systems. (n.d.). ESP32 series datasheet. https://www.espressif.com

Feil-Seifer, D., & Matarić, M. J. (2005). Defining socially assistive robotics. In Proceedings of the 9th International Conference on Rehabilitation Robotics (pp. 465–468). IEEE.

Fraden, J. (2016). Handbook of modern sensors: Physics, designs, and applications (5th ed.). Springer.

Hall, D., & Williams, C. (Directors). (2014). Big Hero 6 [Film]. Walt Disney Animation Studios.

Hawker, G. A., Mian, S., Kendzerska, T., & French, M. (2011). Measures of adult pain. Arthritis Care & Research, 63(S11), S240–S252. https://doi.org/10.1002/acr.20543

International Organization for Standardization. (2017a). Medical electrical equipment — Part 2-56: Particular requirements for basic safety and essential performance of clinical thermometers for body temperature measurement (ISO Standard No. 80601-2-56:2017).

International Organization for Standardization. (2017b). Medical electrical equipment — Part 2-61: Particular requirements for basic safety and essential performance of pulse oximeter equipment (ISO Standard No. 80601-2-61:2017).

Jubran, A. (2015). Pulse oximetry. Critical Care, 19, Article 272. https://doi.org/10.1186/s13054-015-0984-8

Maxim Integrated. (n.d.). MAX30102: High-sensitivity pulse oximeter and heart-rate sensor for wearable health [Data sheet].

Melexis. (n.d.). MLX90614 family: Single and dual zone infra red thermometer in TO-39 [Data sheet].

Mori, M., MacDorman, K. F., & Kageki, N. (2012). The uncanny valley. IEEE Robotics & Automation Magazine, 19(2), 98–100. https://doi.org/10.1109/MRA.2012.2192811

Pu, L., Moyle, W., Jones, C., & Todorovic, M. (2019). The effectiveness of social robots for older adults: A systematic review and meta-analysis of randomized controlled studies. The Gerontologist, 59(1), e37–e51. https://doi.org/10.1093/geront/gny046

Republic Act No. 9994. (2010). Expanded Senior Citizens Act of 2010. Congress of the Philippines.

Rus, D., & Tolley, M. T. (2015). Design, fabrication and control of soft robots. Nature, 521(7553), 467–475. https://doi.org/10.1038/nature14543

Sanan, S., Ornstein, M. H., & Atkeson, C. G. (2011). Physical human interaction for an inflatable manipulator. In 2011 Annual International Conference of the IEEE Engineering in Medicine and Biology Society (pp. 7401–7404). IEEE.

Wada, K., & Shibata, T. (2007). Living with seal robots—Its sociopsychological and physiological influences on the elderly at a care house. IEEE Transactions on Robotics, 23(5), 972–980. https://doi.org/10.1109/TRO.2007.906261

World Health Organization. (2003). Adherence to long-term therapies: Evidence for action.

World Health Organization. (2011). Pulse oximetry training manual. WHO Patient Safety.

World Health Organization. (2021, April 26). Falls [Fact sheet]. https://www.who.int/news-room/fact-sheets/detail/falls

APPENDICES

Appendix A. Materials, Quantity, and Expenses

Prices are estimates from local and online stores in October 2026. Replace them with the amounts on the official receipts.

Table 16

Expense Summary of the Study

Item	Qty	Estimated unit price (PHP)	Subtotal (PHP)
Ripstop nylon 70D, white (per meter)	3	150	450
Seam tape, zipper, thread, felt	1 lot	200	200
12 V DC blower fan with guard	1	250	250
12 V JGA25-370 geared motor with bracket	2	350	700
65-mm wheels with couplers (pair); 2-in caster	1 set	210	210
L298N motor driver	1	120	120
5-channel line sensor; 2 bumper microswitches	1 set	220	220
ESP32 development board	1	380	380
MLX90614 (GY-906) module	1	550	550
MAX30102 module	1	180	180
HC-SR04 ultrasonic sensor	1	70	70
0.96-in OLED display	1	180	180
DFPlayer Mini, 8 GB microSD, 3 W speaker	1 set	380	380
KY-037 sound sensor, button, 10 kΩ dial	1 set	160	160
12 V 7 Ah SLA battery	1	650	650
LM2596, D4184 module, fuse, switch, diode	1 set	180	180
Pendant: ESP32-C3, MPU6050, LiPo, TP4056, buzzer, case	1 set	600	600
Wires, perfboard, chest box, plywood, PVC, screws	1 lot	550	550
Black electrical tape, 19 mm	3 rolls	40	120
Rigid shell for the usability test	1	300	300
70% alcohol, wipes, data sheets	1 lot	150	150
Total (estimated)			**6,600**

Note. Table created by the researchers, 2026.

Appendix B. Raw Data Sheets

Table 17

Raw Data Sheet for Navigation Trials

Trial	Target	Arrived at	Correct (Y/N)	Time (s)	Obstacle	Stop distance (cm)	Notes
1	Chair						
2	Bed						
3	Kitchen						
4	Dock						

Note. Table created by the researchers, 2026.

Table 18

Raw Data Sheet for Health Checks

Code	Trial	ALAGA T (°C)	Ref T (°C)	ALAGA PR	Ref PR	ALAGA SpO_2	Ref SpO_2	Pain	Time (s)
V01	1								
V01	2								
V01	3								
V02	1								
V02	2								
V02	3								

Note. Table created by the researchers, 2026.

Appendix C. Photographs of the Procedure

[NEED: dated photographs of each step in Figure 4, each with a bold label and the note "Photographs taken by the researchers, 2026." No faces of volunteers without written permission.]

Appendix D. Risk Register

Table 19

Risk Register of the Study

Hazard	Risk	Control
Moving robot	Bumping a person or tripping	0.2 m/s top speed; soft body; bumper and ultrasonic stop; tape kept flat
Simulated falls	Bruises or sprains	Researchers only; 20-cm gym mats; pads; adult supervision
Blower fan and wheels	Fingers caught	Finger guard; wheels inside the base outline
Battery and wiring	Short circuit, heat	Inline fuse; closed box; supervised charging
Shared sensors	Spread of germs	Wipe with 70% alcohol between participants
Messages and data	Loss of privacy	Private Telegram group; codes; deleted after the fair

Note. Table created by the researchers, 2026.

Appendix E. Ethics Statement and Informed Consent

This study involved human participants, including older adults. The research plan, consent forms, and data sheets were approved by the school Institutional Review Board on [date] before any testing. Participation was voluntary, and participants could stop at any time without penalty.

Informed consent (summary read to each older adult): “You are invited to try ALAGA, a robot that reminds, checks temperature, pulse, and oxygen, and plays music. You will sit while a sensor reads your forehead from a short distance and you place one finger in a slot. The school clinic's instruments will be used too. You will answer a short questionnaire. It takes about 20 minutes and does not hurt. You will not be asked to fall or to walk with the robot. Your name will not be recorded.”

[NEED: signed consent forms, parental consent and minor assent forms, and the IRB approval form.]

Appendix F. ISEF Forms

[NEED: signed Checklist for Adult Sponsor (1), Student Checklist (1A), Research Plan/Project Summary, Approval Form (1B), Risk Assessment Form (3), Human Participants Form (4) with IRB approval, and informed consent forms.]

Appendix G. Wiring Diagram and Program

Table 20

Pin Connections of ALAGA

Part	Part pin	ESP32 pin	Note
MLX90614, MAX30102, OLED	SDA / SCL	21 / 22	I²C, 100 kHz, 3.3 V
L298N	ENA, IN1, IN2 / ENB, IN3, IN4	14, 18, 19 / 23, 5, 15	Left / right motor
Line sensor (5-channel)	L1–L5	35, 36, 39, 4, 13	3.3 V supply
HC-SR04 (front)	TRIG / ECHO	32 / 33	ECHO through 1 kΩ/2 kΩ divider
Bumper switches	COM / NO	GND / 2	Internal pull-up
Dial, button, sound sensor	wiper / leg / D0	34 / 27 / 26	Button to GND
DFPlayer Mini	TX / RX	16 / 17	1 kΩ on RX
D4184 MOSFET module	PWM	25	Switches the 12 V fan

Note. Table created by the researchers, 2026.

The programs are in the project folder: firmware/alaga/alaga.ino for the robot (ESP32 Dev Module) and firmware/alaga_pendant/alaga_pendant.ino for the pendant (ESP32-C3). The build guide lists the libraries, the Telegram setup, and the 20 voice prompts.

[NEED: photograph or drawing of the final wiring.]

Appendix H. Curriculum Vitae

[NEED: one-page curriculum vitae of each researcher with a 2 × 2 photo.]